

lecture 06

bayes

Bayesian inference; rejection algorithm; a Bayesian invariance property

Paper and Workshop

review

example exam questions

Define a *first difference*. Explain how and why the first difference unifies diverse statistical models. Use two diverse examples. Use words and/or equations, but be precise.

Explain the conceptual framework of {marginaleffects}. Identify the main functions and their key arguments.

For claims about the effect of X on Y , what is a *good default* quantity of interest? Defend your choices.

Bayesian Inference

Frequentism: a toolkit for evaluating estimators (e.g., bias, efficiency, coverage); ML offers some frequentist guarantees.

Bayesian inference: a method for finding the “correct” inference.

Bayesian Inference

(just Bayes' rule)



Bayesian Inference

(just Bayes' rule)



For continuous random variables [\[edit\]](#)

For two continuous [random variables](#) X and Y , Bayes' theorem may be analogously derived from the definition of [conditional density](#):

$$f_{X|Y=y}(x) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

$$f_{Y|X=x}(y) = \frac{f_{X,Y}(x,y)}{f_X(x)}$$

Therefore,

$$f_{X|Y=y}(x) = \frac{f_{Y|X=x}(y)f_X(x)}{f_Y(y)}.$$

IMPORTANT:
we have 3 different fs.

$$f(\theta | x) = \frac{f(x | \theta) \times f(\theta)}{\int_{-\infty}^{\infty} f(x | \theta) f(\theta) d\theta}$$

$$\boxed{f(\theta | x)} = \frac{\boxed{f(x | \theta)} \otimes \boxed{f(\theta)}}{\int_{-\infty}^{\infty} f(x | \theta) f(\theta) d\theta}$$

1. Choose a distribution for the data (i.e., “**the likelihood**” or model of the data).
2. Choose a **prior distribution** to describe your prior beliefs.
3. Update the prior distribution upon observing the data by computing the **posterior distribution**.
 - (i) Multiply the likelihood and prior density.
 - (ii) Normalize, so that posterior integrates to one.

easy!

HARD!

$$f(\theta \mid x) = \frac{f(x \mid \theta) \times f(\theta)}{\int_{-\infty}^{\infty} f(x \mid \theta) f(\theta) d\theta}$$

HARD!

Solutions

Conjugate Prior

Choose a likelihood and prior so that the posterior and prior are from the same family of probability distributions (e.g., both beta).
(restricts prior specification)

Unnormalized Posterior

$$f(\theta \mid x) \propto f(x \mid \theta) \times f(\theta)$$

(correct shape, but doesn't integrate to one)

Unnormalized Log-Posterior

$$\log f(\theta \mid x) \propto \log f(x \mid \theta) + \log f(\theta)$$

(turns out to be really handy!)

Simple Example

toothpaste cap problem

The Likelihood

$$f(x_i \mid \pi) = \pi^{x_i} (1 - \pi)^{(1-x_i)}$$

$$f(x \mid \pi) = f(x_1 \mid \pi) \times \dots \times f(x_N \mid \pi) = \prod_{i=1}^N f(x_i \mid \pi)$$

$$f(x \mid \pi) = \pi^k (1 - \pi)^{(N-k)}, \text{ where } k = \sum_{i=1}^N x_i$$

Recall we had $N = 150$ and $k = 8$
as an example dataset—we've got
the likelihood already.

The Prior

the beta distribution

$$f(x) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}$$

the prior
(which is beta)

$$f(\pi) = \frac{1}{B(\alpha^*, \beta^*)} \pi^{\alpha^*-1} (1-\pi)^{\beta^*-1}$$

two prior parameters

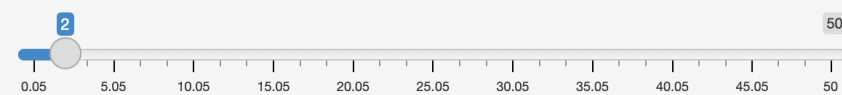
What prior parameters make sense
for the toothpaste cap problem?

<https://carlislerainey.shinyapps.io/shiny-beta/>

Explore the Beta Distribution

Shape Parameters

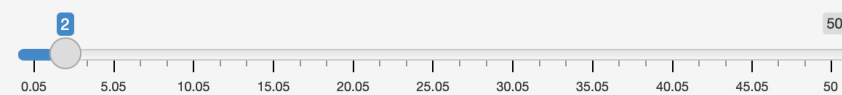
α ?



Alpha (exact)

2

β ?



Beta (exact)

2

Presets

- Uniform (1, 1)
- Right-skew (2, 10)
- Mound (20, 20)
- U-shape (0.5, 0.5)
- Left-skew (10, 2)

Set α , β from Targets

Enter Mean and SD, then click Preview.

Mean

0.5

SD

0.1

Preview

Click Apply to use these values of α and β .

Apply

Save Current Plot (PNG)

PDF

CDF

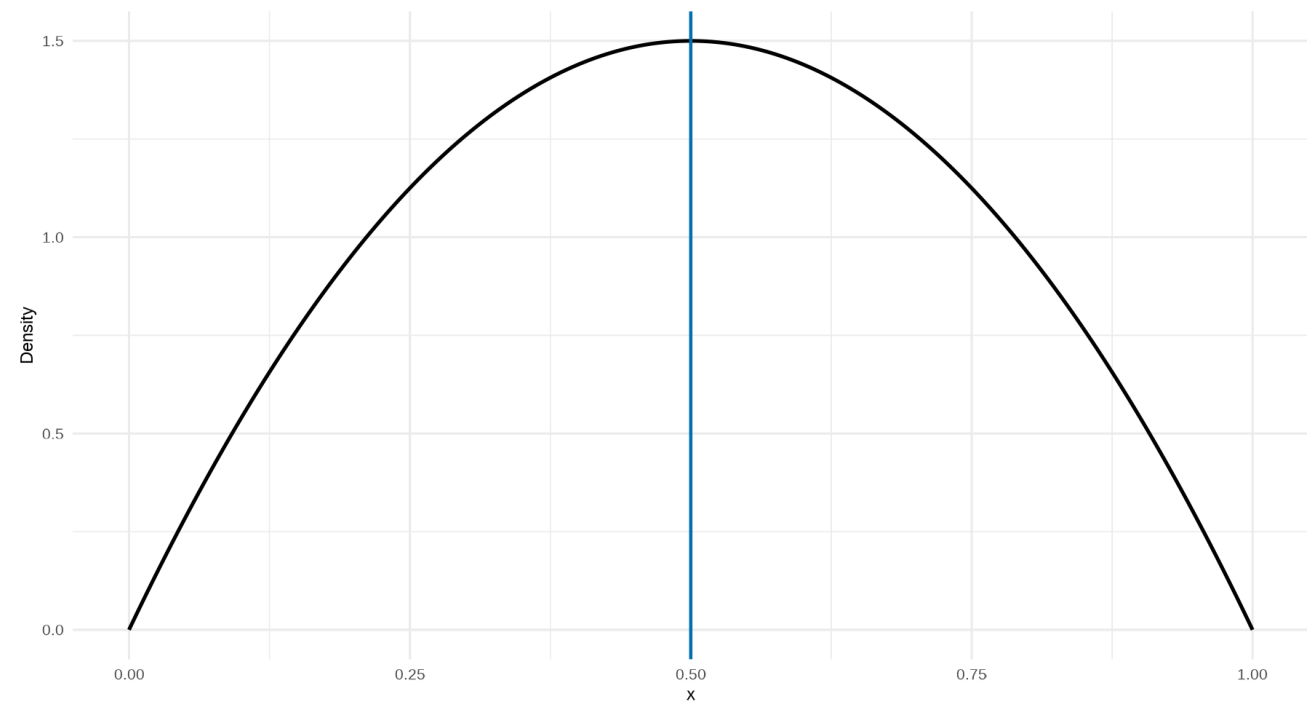
Simulations

Summaries

Learn

Beta PDF

$\alpha=2.00$, $\beta=2.00$ • Mean=0.50 • SD=0.22



Understand your priors.

<https://gist.github.com/carlislerainey/45414e0d9f22e4e1960449402e6a8048>

Let's plot our priors.

<https://gist.github.com/carlislerainey/45414e0d9f22e4e1960449402e6a8048>

(note: this code plots priors and posteriors—only run the prior part for now)

The Posterior

$$f(\theta \mid x) = \frac{f(x \mid \theta) \times f(\theta)}{\int_{-\infty}^{\infty} f(x \mid \theta) f(\theta) d\theta}$$

$$f(\pi \mid x) = \frac{\left[\pi^k (1 - \pi)^{(N-k)} \right] \times \left[\frac{1}{B(\alpha^*, \beta^*)} \pi^{\alpha^* - 1} (1 - \pi)^{\beta^* - 1} \right]}{C_1}$$

work on whiteboard

Some useful equations below

the posterior (where we're starting)

$$f(\pi | x) = \frac{\left[\pi^k (1 - \pi)^{(N-k)} \right] \times \left[\frac{1}{B(\alpha^*, \beta^*)} \pi^{\alpha^* - 1} (1 - \pi)^{\beta^* - 1} \right]}{C_1}$$

the posterior (where we're heading)

$$\begin{aligned} f(\pi | x) &= \frac{\overbrace{\pi^{(\alpha^* + k) - 1}}^{\alpha'} \times (1 - \pi)^{\overbrace{[\beta^* + (N - k)] - 1}^{\beta'}}}{B(\alpha^* + k, \beta^* + [N - k])} \\ &= \frac{\pi^{\alpha' - 1} \times (1 - \pi)^{\beta' - 1}}{B(\alpha', \beta')}, \text{ where } \alpha' = \alpha^* + k \text{ and } \beta' = \beta^* + [N - k] \end{aligned}$$

the beta distribution

$$f(x) = \frac{1}{B(\alpha, \beta)} x^{\alpha - 1} (1 - x)^{\beta - 1}$$

prior

$\alpha_{\text{star}} \leftarrow 3$

$\beta_{\text{star}} \leftarrow 15$

data

$N \leftarrow 150$

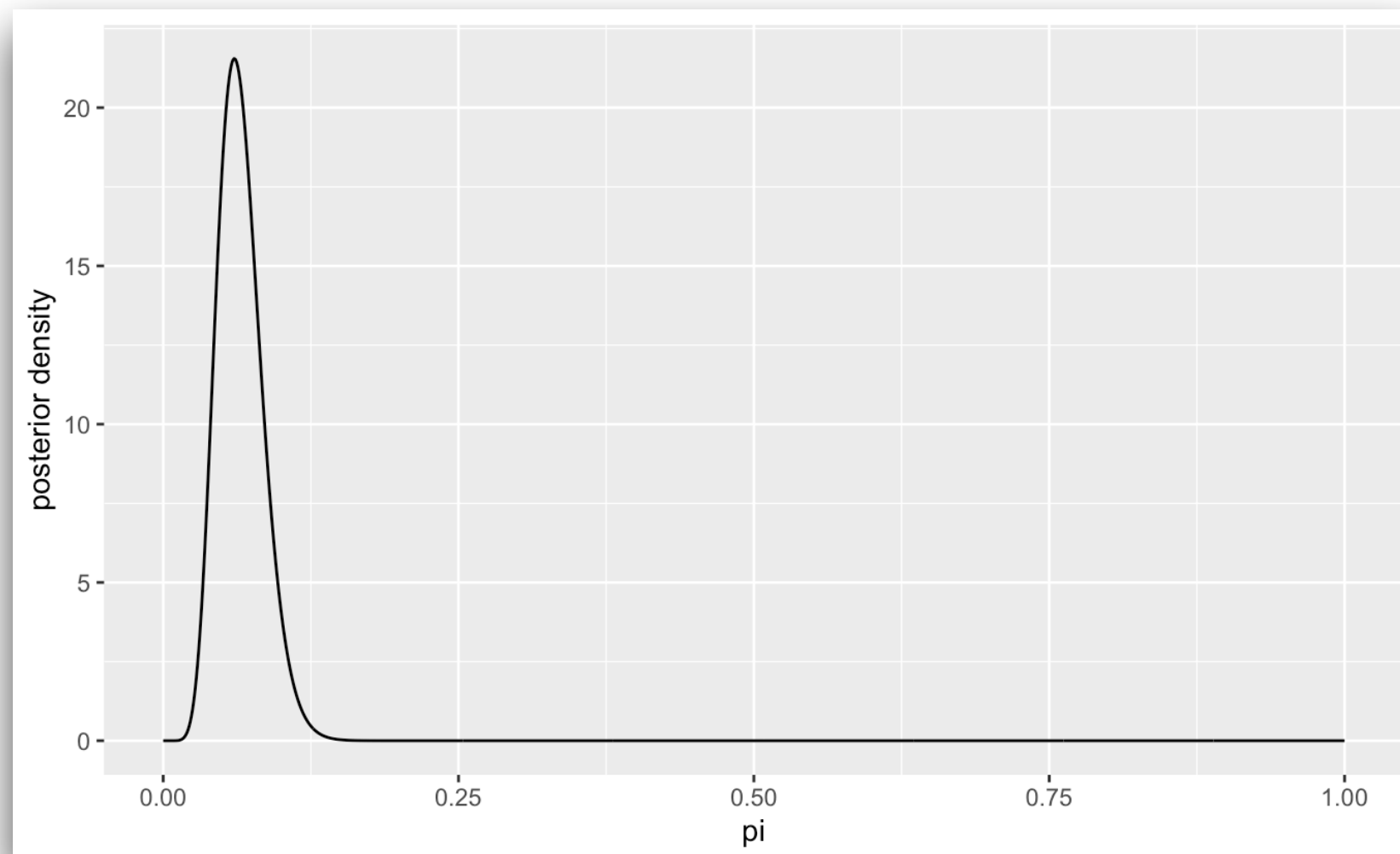
$k \leftarrow 8$

posterior

$\alpha_{\text{prime}} \leftarrow \alpha_{\text{star}} + k$

$\beta_{\text{prime}} \leftarrow \beta_{\text{star}} + (N - k)$

```
# plot posterior pdf
ggplot() +
  xlim(0, 1) +
  stat_function(fun = dbeta, n = 1001,
               args = list(shape1 = alpha_prime,
                           shape2 = beta_prime)) +
  labs(x = "pi",
       y = "posterior density")
```



```
# find posterior mean  
alpha_prime/(alpha_prime + beta_prime)
```

```
> # find posterior mean  
> alpha_prime/(alpha_prime + beta_prime)  
[1] 0.06547619  
> |
```

Let's try it with **your** prior

<https://gist.github.com/carlislerainey/5064ecde24ea764817da73b037e7a18b>

Let's compare our inferences

<https://gist.github.com/carlislerainey/155e2fee1c01620736120c9a8bbccd6d>

A Few Conjugate Priors

Model of Data

Prior

Bernoulli

Beta

Poisson

Gamma

Normal

Normal; Inverse Gamma

Exponential

Gamma

Poisson: in the notes

exponential: an exercise

Rejection Algorithm

Observation: simulations from the posterior are as good as a closed-form solution.

Implication: we don't need to find the posterior; we need to find a way to *simulate from* the posterior.

```
# prior parameters
alpha_prior <- 3
beta_prior <- 15

# data
k <- 8
N <- 150

# posterior parameters
alpha_posterior <- alpha_prior + k
beta_posterior <- beta_prior + N - k

# for compact calculations below
a1 <- alpha_posterior
b1 <- beta_posterior

# posterior simulation; trivial
n_sims <- 100000
pi_tilde <- rbeta(n_sims, shape1 = a1, shape2 = b1)

# posterior mean
a1 / (a1 + b1) # closed form
```

```
[1] 0.06547619
```

```
mean(pi_tilde) # posterior sim
```

```
[1] 0.06543861
```

```
# posterior sd  
sqrt((a1 * b1) / ((a1 + b1)^2 * (a1 + b1 + 1))) # closed form
```

```
[1] 0.01902802
```

```
sd(pi_tilde) # posterior sim
```

```
[1] 0.01901066
```

```
# posterior 5th percentile  
qbeta(0.05, shape1 = a1, shape2 = b1) # closed form
```

```
[1] 0.03737493
```

```
quantile(pi_tilde, probs = 0.05) # posterior sims
```

```
5%  
0.03742531
```

```
# posterior 95th percentile  
qbeta(0.95, shape1 = a1, shape2 = b1) # closed form
```

```
[1] 0.09945329
```

```
quantile(pi_tilde, probs = 0.95) # posterior sims
```

```
95%  
0.09936871
```

But this idea is

even

more

powerful

than it initially seems

Surprising fact: It is easier to sample from a distribution than find the closed form.

How it is even possible to sample from distribution without knowing the closed form?



Example Algorithm: Rejection

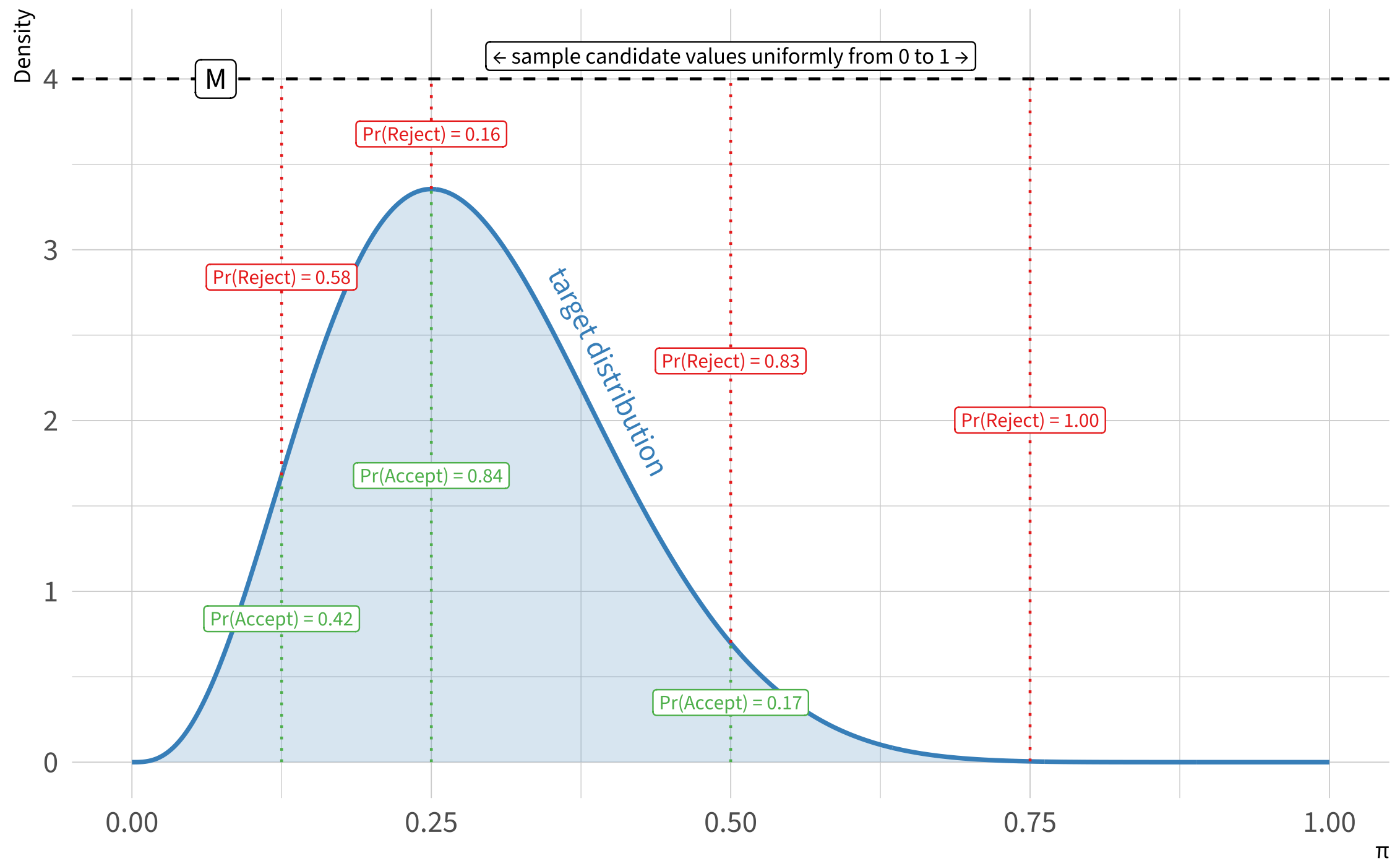
Example Targets:

Beta(4, 10)

Weird One! (Sawtooth Prior)

Illustrating the logic of the rejection algorithm

For a beta(4, 10) target distribution.



Algorithm: Rejection Sampling^a

Inputs:

- The unnormalized posterior $f(\pi \mid y)$ on $[0, 1]$.
- Desired number of draws S .
- Envelope constant M such that $M > f(\pi) \forall \pi$.

Algorithm:

1. **Initialize:** $Sets=1$.

2. **Repeat until** $s = S$:

(a) $z \sim \text{Uniform}(0, 1)$.

(b) $u \sim \text{Uniform}(0, 1)$.

(c) **Accept–reject step:**

If $u \leq f(z)/M$, accept: set $\pi^{(s)} = z$, $s \leftarrow s + 1$.

Otherwise, reject z and return to Step 2a.

Output: $\pi^{(1)}, \pi^{(2)}, \dots, \pi^{(S)} \sim f(\pi \mid y)$.

^aTo make the rejection algorithm simple, I've written it to apply specifically to the posterior for the Bernoulli model, which has support $[0, 1]$. The target density doesn't need to be a posterior and can have support other than $[0, 1]$. The proposal distribution doesn't have to be uniform. The key is that M is larger than the maximum of the target distribution and draws are accepted with probability $f(z)/M$.

```

rej <- function(f, S, M) {

  # create containers and initialize counters
  samples <- numeric(S) # container to store samples
  rejects <- NULL # container to track rejected values; for teaching; slow!
  s <- 1 # currently trying to take sample 1
  n_prop <- 0 # count proposals (for an acceptance-rate message)

  # so long as the current sample s is less
  #   than the desired samples S.
  #   do the following:
  while (s <= S) {

    # A: propose  $z \sim \text{uniform}(0,1)$ 
    z <- runif(1)

    # B: draw  $u \sim \text{uniform}(0,1)$ 
    u <- runif(1)

    # C: Accept or reject
    fz <- f(z) # compute once, for efficiency

    ## scenario 1:  $u \leq f(z)/M \rightarrow \text{Accept}$ 
    if (u <= fz / M) {
      samples[s] <- z
      s <- s + 1
    }

    ## scenario 2:  $f(z) > M \rightarrow \text{shouldn't happen; error}$ 
    if (fz > M) stop("Stop: Envelope M is too small.") # find appropriate M

    ## scenario 3:  $u > f(z)/M \rightarrow \text{Reject}$ 
    ##   tracking these values just for teaching and learning--not needed usually
    if (u > fz / M) {
      rejects <- c(rejects, z)
    }

    # track total proposals so far
    n_prop <- n_prop + 1
  }

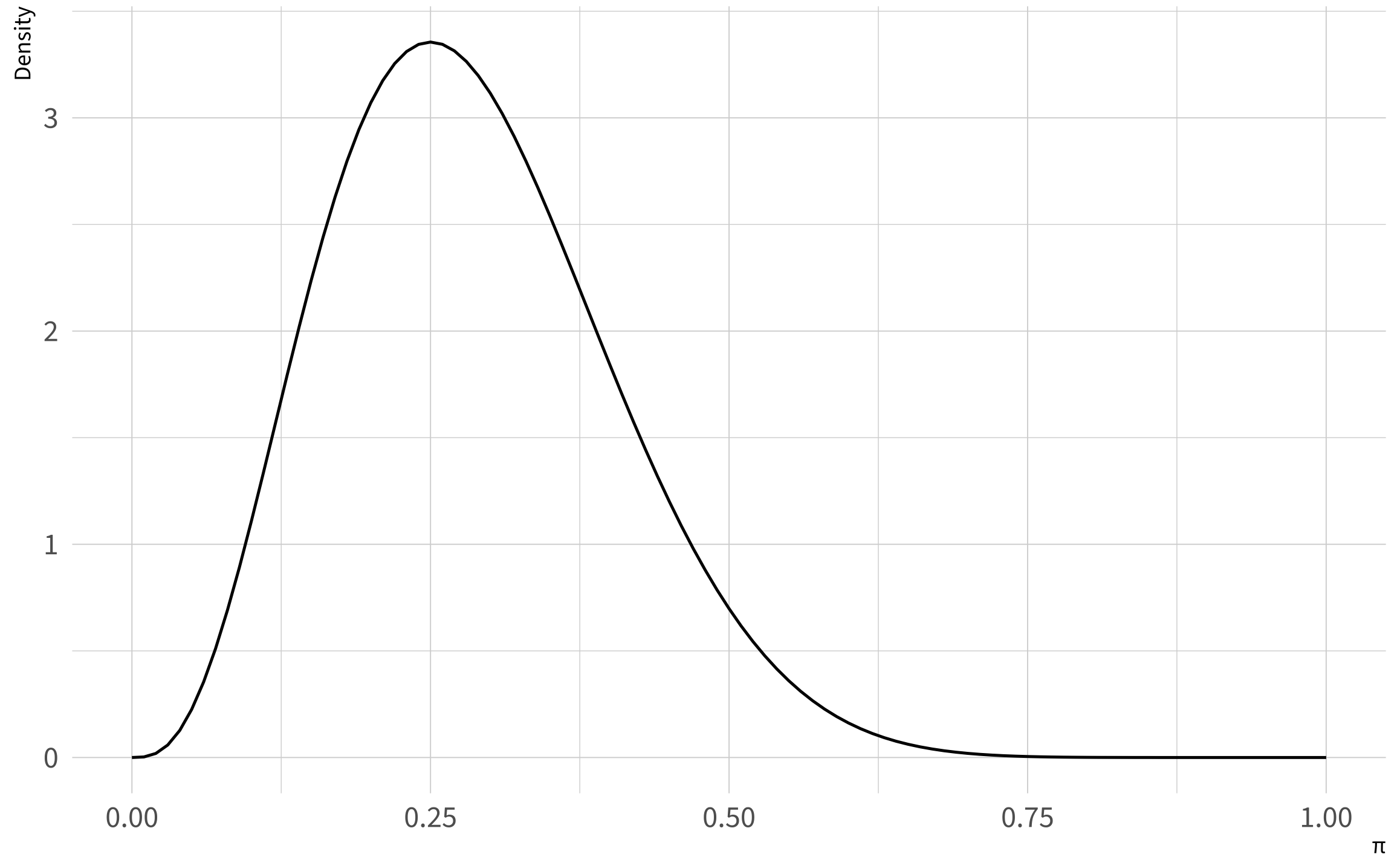
  # return
  list(
    n_prop = n_prop,
    acc_rate = S / n_prop,
    samples = samples,
    rejects = rejects
  )
}

```

scenario 1 is the key

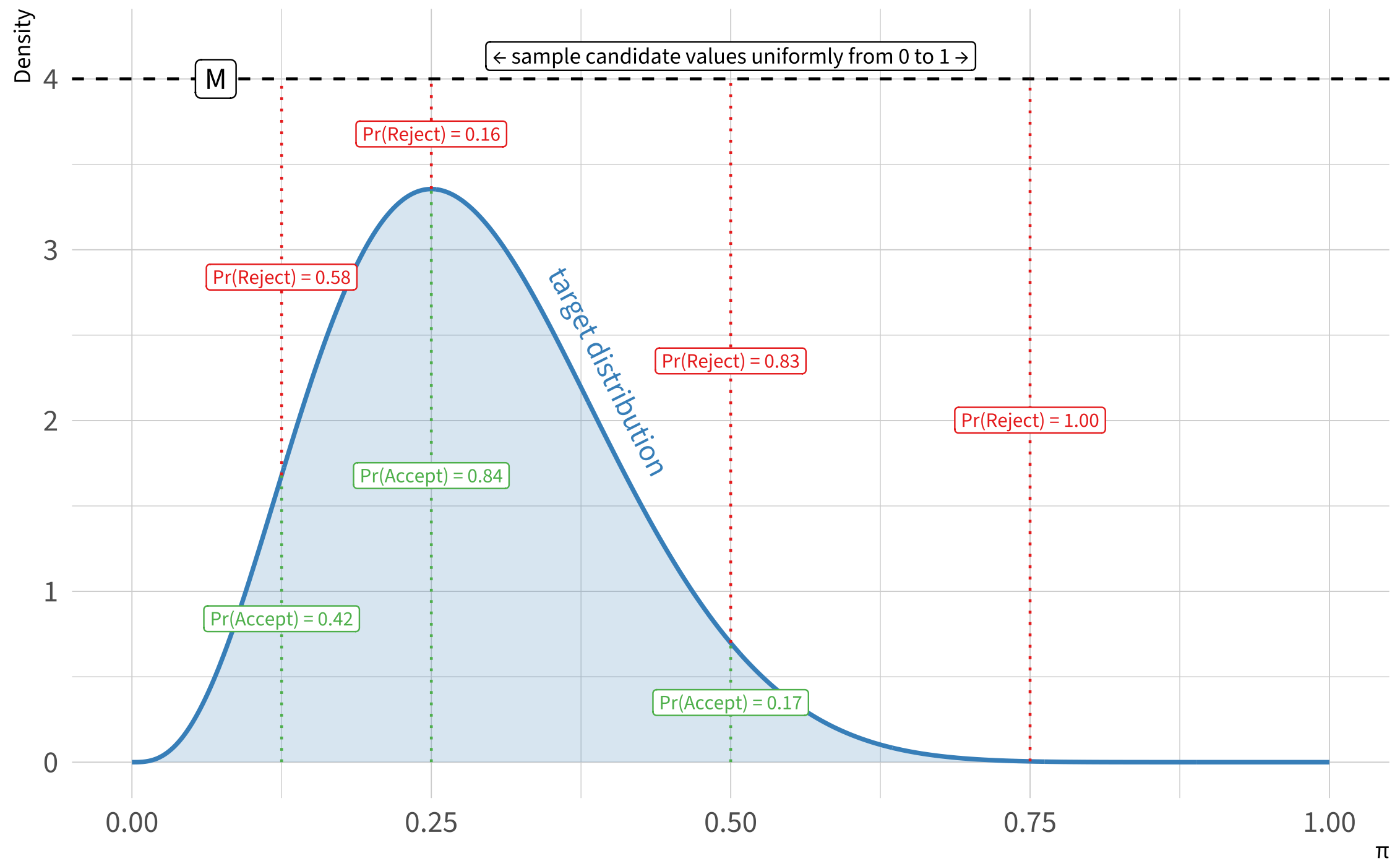
Target distribution

beta(4, 10)



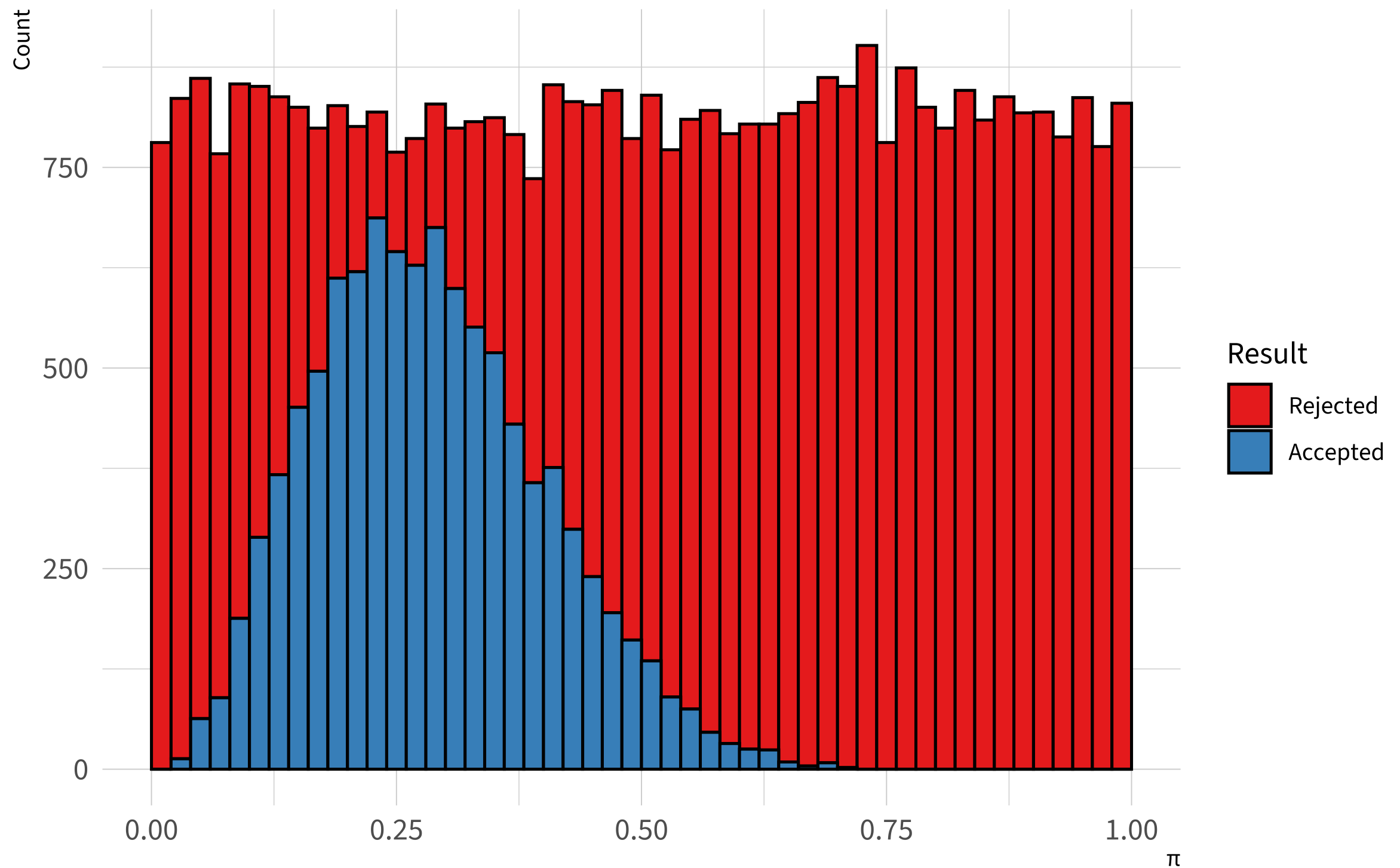
Illustrating the logic of the rejection algorithm

For a beta(4, 10) target distribution.



Samples from rejection algorithm

10,000 accepted samples; 30,870 rejected values



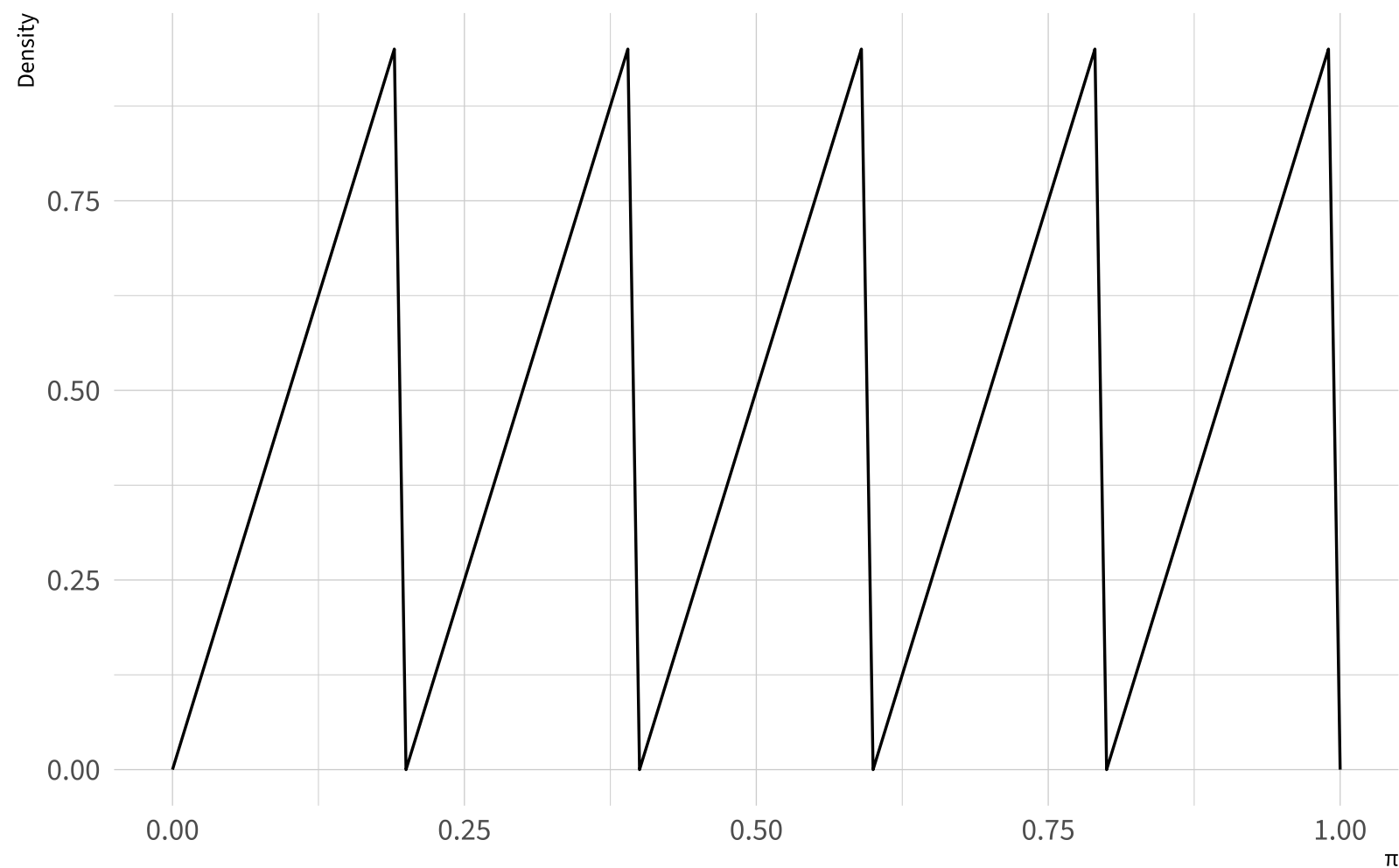
```
> # posterior mean  
> 4/(4 + 10) # closed-form  
[1] 0.2857143  
> mean(r$samples) # simulation  
[1] 0.2847601
```

Let's make it weird

```
# unnormalized prior
prior_saw <- function(p, n_teeth = 5) {
  ((n_teeth*p) %% 1)
}
```

Sawtooth "prior" that does not integrate to one

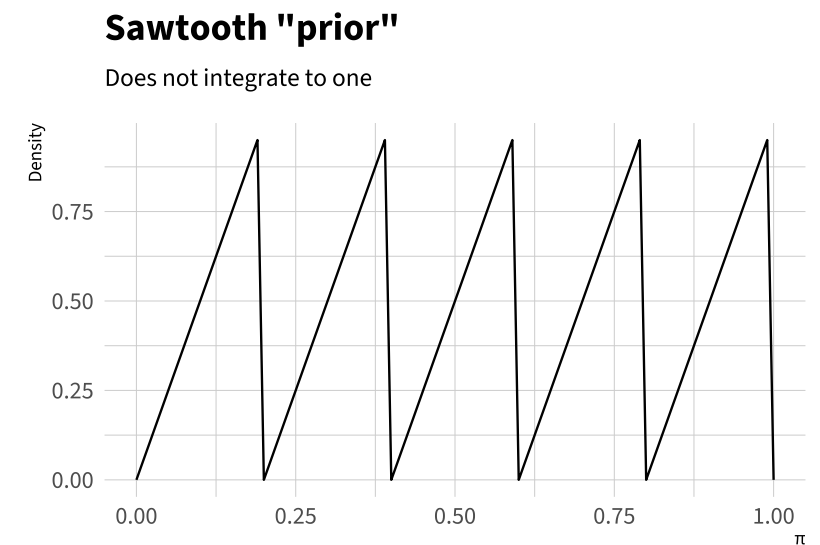
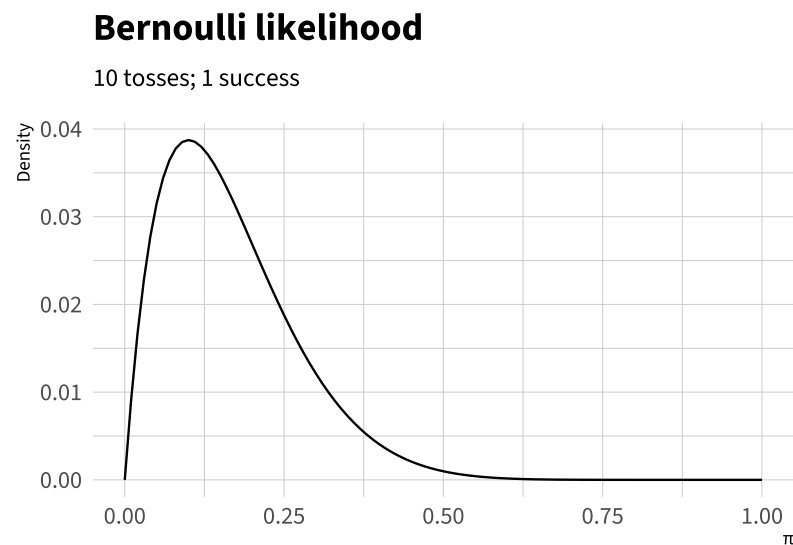
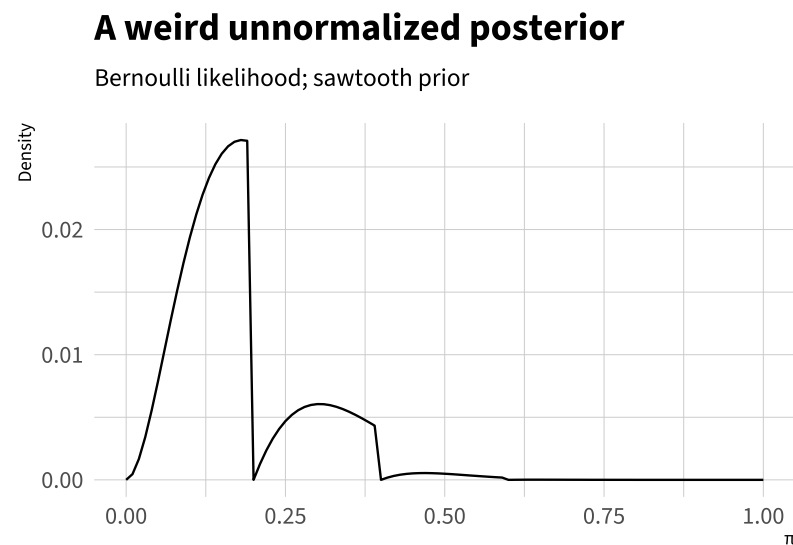
A weird example to illustrate



$$\text{posterior} = \frac{\text{Bernoulli likelihood} \times \frac{\text{sawtooth prior}}{\text{normalizing constant}}}{\text{normalizing constant}}$$

$$\text{posterior} \propto \text{Bernoulli likelihood} \times \text{unnormalized sawtooth prior}$$

posterior $\propto$ Bernoulli likelihood $\times$ unnormalized sawtooth prior



Find mean of posterior?!?
Find quantiles of posterior?!?
HOW?!?

```
# unnormalized prior
prior_saw <- function(p, n_teeth = 5) {
  ((n_teeth*p) %% 1)
}
```

```
# likelihood (10 tosses; 1 success )
lik <- function(p) {
  p^1 * (1-p)^9
}
```

```
# unnormalized posterior
unnormalized_posterior <- function(p) {
  lik(p)*prior_saw(p)
}
```

```
# rejection algorithm
r <- rej(unnormalized_posterior, S = 10000, M = 0.03)
```

```
# posterior mean
mean(r$samples)
```

```
> # posterior mean
> mean(r$samples)
[1] 0.1759005
```

```
# 90% credible interval
quantile(r$samples, probs = c(0.05, 0.95))
```

```
# histogram
hist(r$samples, breaks = 100)
```

```
> r <- rej(unnormalized_posterior, S = 10000, M = 0.03)
```

👉 Successfully generated 10,000 samples! 🎉

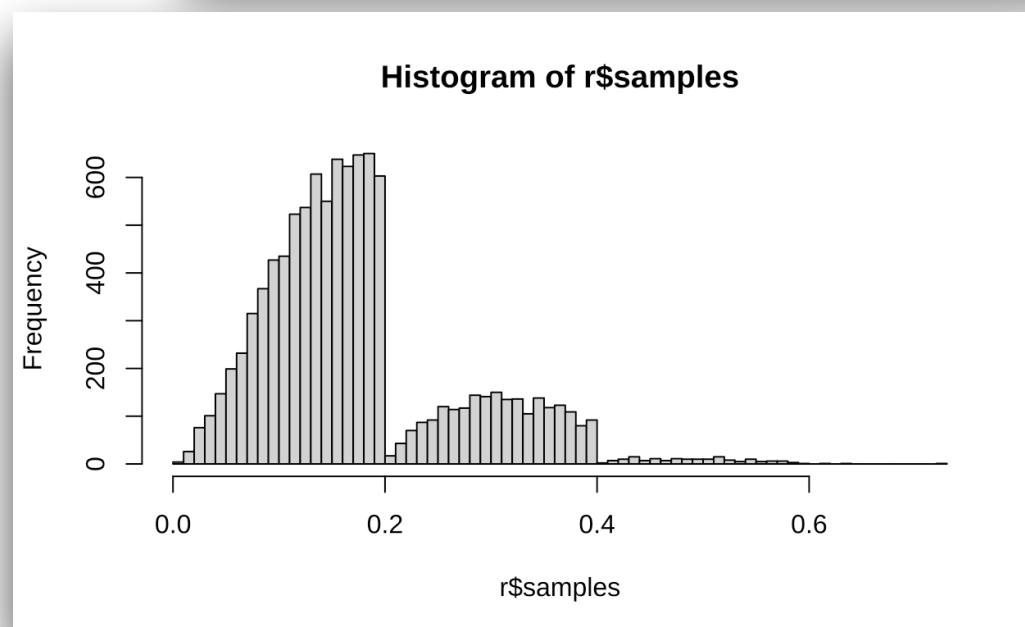
✅ Accepted samples: 10,000

❌ Rejected samples: 58,263

% Acceptance rate: 15%

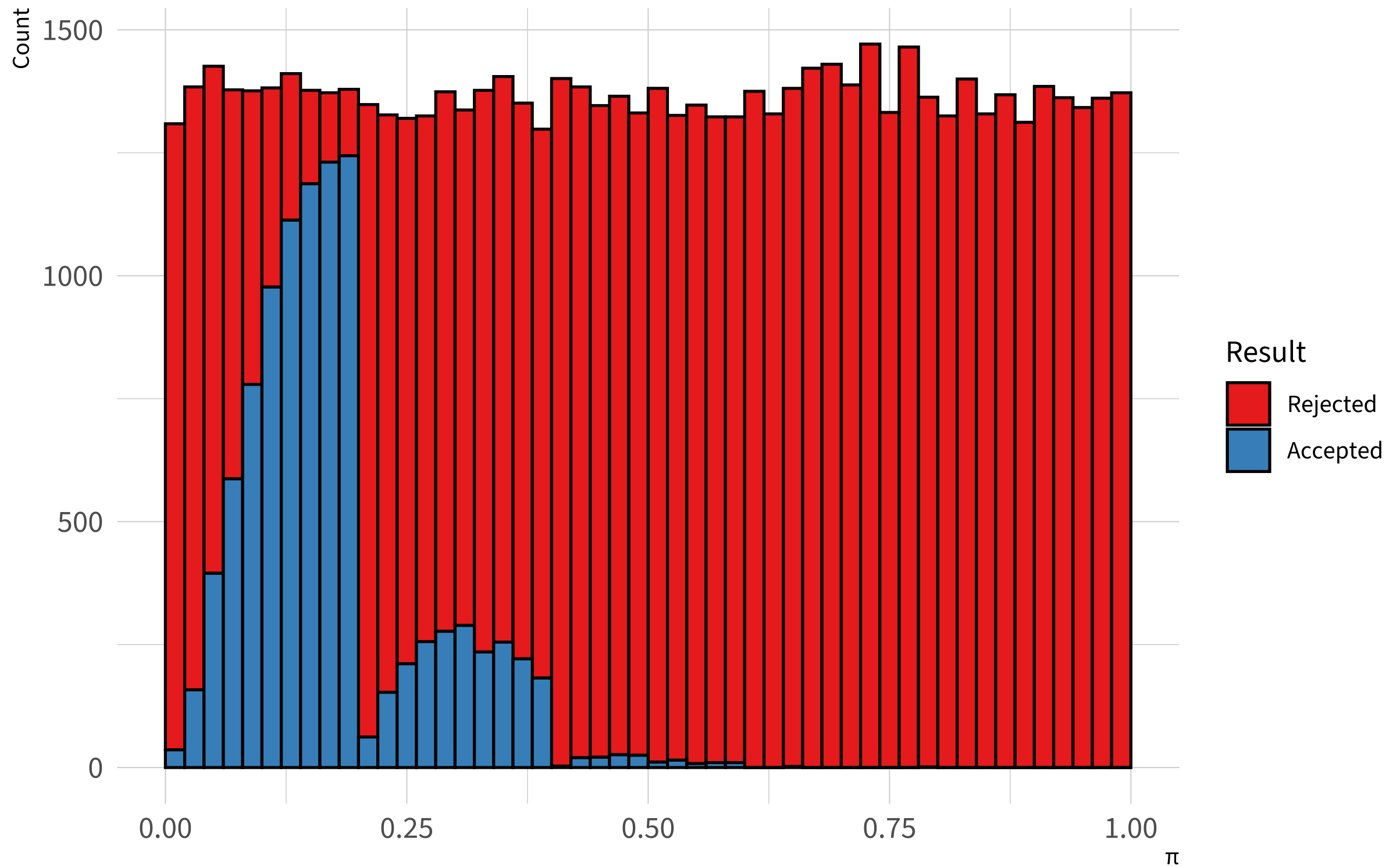
🕒 Total time: 7.4s

```
> # 90% credible interval
> quantile(r$samples, probs = c(0.05, 0.95))
               5%      95%
0.05770977 0.36608499
```



Samples from rejection algorithm

10,000 accepted samples; 58,295 rejected values



$$f(\theta \mid x) = \frac{f(x \mid \theta) \times f(\theta)}{\int_{-\infty}^{\infty} f(x \mid \theta) f(\theta) d\theta}$$

HARD!

Notice how the rejection algorithm allowed us to completely skip the hard parts.

Bayesian Normal Linear Regression (Conjugate Prior)

$$\text{Likelihood: } y \mid \beta, \sigma^2 \sim \mathcal{N}(X\beta, \sigma^2 I)$$

$$\text{Prior: } \beta \mid \sigma^2 \sim \mathcal{N}(\beta^*, \sigma^2 V^*), \quad \sigma^2 \sim \text{Inv-Gamma}(a^*, b^*)$$

$$\text{Posterior: } \beta \mid \sigma^2, y \sim \mathcal{N}(\beta', \sigma^2 V'), \quad \sigma^2 \mid y \sim \text{Inv-Gamma}(a', b')$$

$$\text{Updates: } V' = ((V^*)^{-1} + X^\top X)^{-1}, \quad \beta' = V'((V^*)^{-1}\beta^* + X^\top y)$$

$$a' = a^* + \frac{n}{2}, \quad b' = b^* + \frac{1}{2} (y^\top y + (\beta^*)^\top (V^*)^{-1} \beta^* - (\beta')^\top (V')^{-1} \beta')$$

Parameters: σ^2 = error variance, V^*, V' = prior/posterior covariances, a^*, a' = shape, b^*, b' = scale.

Note: $\beta \mid y \sim t_{2a'}\left(\beta', \frac{b'}{a'} V'\right)$ (marginal posterior after integrating out σ^2).

A Bayesian Invariance Property

$$\hat{\tau}_{mean} \neq \tau \left(\hat{\theta}_{mean} \right)$$

The transformed posterior mean is *NOT* the
posterior mean of the transformation.

A Bayesian invariance property. Suppose $\{\tilde{\theta}^{(s)}\}_{s=1}^S$ are posterior simulations of θ . Let $\tau = \tau(\theta)$ be a quantity of interest for any function τ . Then posterior simulations of τ can be obtained by applying τ to each draw $\tilde{\theta}^{(s)}$ so that $\tilde{\tau}^{(s)} = \tau(\tilde{\theta}^{(s)})$. Summaries of the posterior distribution of τ (e.g., mean, median, credible intervals) are obtained by summarizing the transformed draws $\{\tilde{\tau}^{(s)}\}_{s=1}^S$.

Transform simulations *before* summarizing.

```
# posterior simulation
```

```
pi_tilde <- rbeta(10000, shape1 = alpha_prime, shape2 = beta_prime)
```

```
oof_tilde <- (1 - pi_tilde)/pi_tilde
```

```
mean(oof_tilde) # correct
```

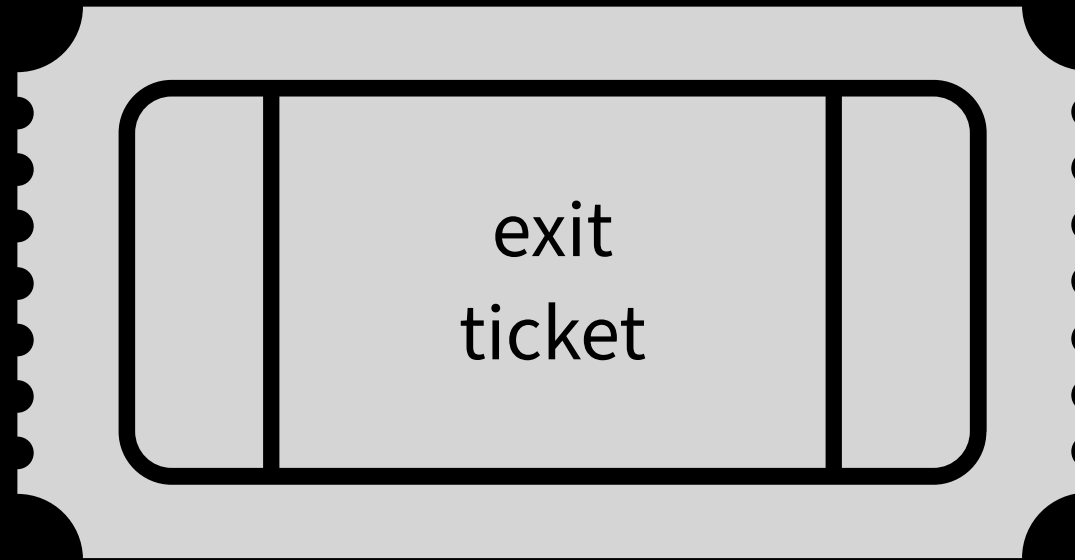
```
(1 - mean(pi_tilde))/mean(pi_tilde) # NOT correct
```

```
> mean(oof_tilde) # correct
```

```
[1] 15.56664
```

```
> (1 - mean(pi_tilde))/mean(pi_tilde) # NOT correct
```

```
[1] 14.15623
```



List three important ideas from today's class. For each, briefly connect it to one or more ideas from last week.